CCY2001: Introduction to Cybersecurity

Lecture 2 – Cybersecurity Basics

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Lecture Outline

What is Cybersecurity?

Definition

"The protection of your software, hardware, and data resources connected and stored on the Internet"

Key Components:

- Protection of software, hardware, and data online
- Ensures Confidentiality, Integrity, and Availability
- Covers physical access, intrusion control, encryption, and privacy
- Involves people, processes, and technology

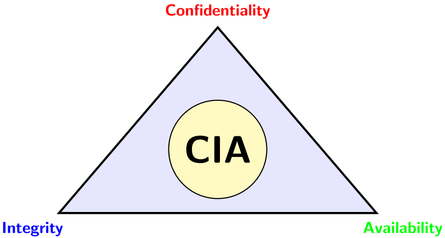
Continuous Process

Awareness → Planning → Implementation → Monitoring → Evaluation

The CIA Triad

The CIA triad is a fundamental model in information security

- **Confidentiality:** Protecting information from unauthorized access and disclosure
- **Integrity:** Ensuring information is accurate, complete, and protected from unauthorized modification
- **Availability:** Guaranteeing that authorized users can access information and systems when needed



Key Point

The three core components work together to guide an organization's security procedures and policies.

Cybersecurity Domains

Why are these domains critical?

- Software resources are vital for business, government, defense, and daily life
- Their CIA is critical for smooth operations
- Physical security and IoT automation are now tied to cybersecurity
- Any physical breach can create serious cyber risks

Critical Domains:

1. Personal data
2. Financial data
3. Commercial data
4. Business critical information
5. Operational continuity
6. Data integrity
7. Availability of online services

Key Point

Cybersecurity is no longer just about digital assets—it encompasses physical security too!

Cybersecurity Elements Classification

Cybersecurity is a vast field that encompasses human behaviors and technological procedures impacting security.

Network Security (NS)	**Leadership Commitment**
Information Security (IS)	**Operational Security (OPSEC)**
Application Security (AS)	**End-user Education**

- **NS:** Protecting networks from unauthorized access, misuse, or attacks
- **IS:** Ensuring CIA of data
- **AS:** Protecting software/apps against vulnerabilities

Cybersecurity Elements Classification (cont.)

Beyond Technical Controls:

Leadership Commitment

- Management support crucial
- Enforcement of cybersecurity policies
- Budget allocation
- Culture of security

End-user Education

- Training users on threats
- Recognizing phishing attempts
- Best security practices
- Human firewall concept

Operational Security (OPSEC)

- Protecting sensitive activities
- Process security
- Preventing exploitation
- Secure operations

Remember

Humans are often the weakest link!

Education and awareness are critical

Importance of Cybersecurity

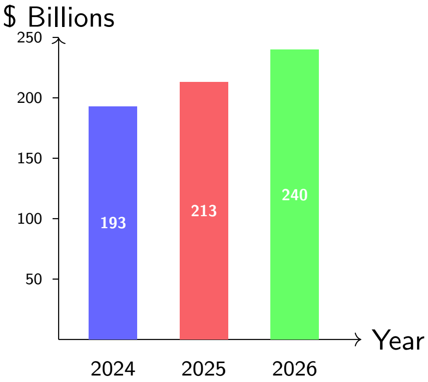
Market Growth and Financial Impact

Global Cybersecurity Spending

- **2024:** \$193 billion
- **2025:** \$213 billion (projected)
- **2026:** \$240 billion (forecasted)

Cost of Data Breaches

Average global cost: **\$4.44 million**



Key Takeaway

Compliance with security standards = key to reducing risk

Introduction to Cyber Attacks

Definition

A **cyberattack** is a malicious attempt to gain unauthorized access to computer systems, networks, or digital devices to:

- Steal, alter, destroy, or expose data
- Disrupt or disable systems

Who Conducts Attacks?

- Individuals or organizations
- Cybercriminals or hackers
- Motivated by:
  - Financial gain
  - Political reasons
  - Revenge
  - Espionage

Common Targets:

- Data servers
- Application servers
- Storage servers
- Financial information
- Operational systems
- Computer networks

Common Types of Cyberattacks

Understanding the Threat Landscape

- Malware Attacks
- Phishing
- DoS/DDoS Attacks
- Man-in-the-Middle
- SQL Injection
- Password Attacks
- Ransomware
- Zero-Day Exploits

Important

These attacks often target different aspects of the CIA Triad!

Cyber Crimes: Specific Examples

Types of Criminal Activities in Cyberspace

1. **Email and Internet Fraud**
   Deceptive messages or websites trick users into giving up sensitive data or money
2. **Identity Fraud**
   Stolen personal details used to impersonate someone for financial or illegal gain
3. **Theft of Financial or Card Data**
   Hackers steal credit/debit card or banking details for fraud
4. **Theft and Sale of Corporate Data**
   Confidential business information stolen and sold to competitors or criminals

Cyber Crimes: Specific Examples (cont.)

5. **Cyberextortion**
   Criminals demand payment to avoid or stop an attack
6. **Ransomware Attacks**
   Malicious software locks systems/data until a ransom is paid
7. **Cryptojacking**
   Hackers secretly use others' computers to mine cryptocurrency

Growing Threat

Cybercrime is expected to skyrocket in coming years, with estimated global costs reaching **trillions of dollars annually**

Objectives of Cyberattacks

What do attackers want to achieve?

Financial Objectives

- Achieving monetary gains
- Stealing money or valuable data
- Cryptocurrency theft

Competitive Objectives

- Damaging brand value
- Stealing business secrets
- Growth hacking email campaigns

Political/Strategic

- Cyber-terrorism
- Stealing government secrets
- Warfare cyber attacks

Remember

Nation-state-sponsored attacks target infrastructure or defense systems

Technical Objectives: CIA Triad Breaches

A successful cyberattack typically compromises one or more elements of the CIA triad

Confidentiality Breach

Sensitive data exposed without authorization

Availability Breach

Legitimate users cannot access systems

Integrity Breach

Data altered or corrupted without authorization

Next Slides

We'll explore each breach type in detail...

Confidentiality Breach: Overview

Definition

Happens when **sensitive data is exposed without authorization**

Examples of Confidential Data:

- Personal identity (SSN, passport, ID)
- Financial info (credit cards, bank details)
- Business secrets & intellectual property
- Personal health information
- Trade secrets

Common Causes:

- Hacking, malware, phishing
- Insider threats or employee negligence
- Lost/stolen devices
- Unattended computers
- Unauthorized access

Confidentiality Breach: Details

Characteristics and Impact

- Can be intentional or unintentional
- Occurs through hacking, negligence, or insider misuse
- Leads to financial losses and misuse of personal/client data
- Many countries have laws and regulations against unauthorized disclosure

Main Sources of Confidentiality Breach

1. Theft or loss of devices (e.g., laptops)
2. Unattended computers with sensitive information
3. Unauthorized access (internal or external)
4. Malware attacks
5. Employees violating confidentiality agreements
6. Using confidential data for personal or business gain

Integrity Breach

Definition

When **data is altered, corrupted, or manipulated without authorization**

Key Principles:

- Data stored on servers should remain accurate, consistent, and valid
- Data is normally stored and transported in encrypted forms
- Maintains confidentiality during storage and transmission

Threats to Data Integrity:

1. Malware on the server
2. Unauthorized modifications
3. Malicious insiders
4. Undoable malicious encryption
5. Manipulation of original data

Important Note

- **Malware** = the big umbrella
- **Virus** = one kind of malware (spreads via files)
- **Ransomware** = malware that locks data and demands payment

Availability Breach

Definition

Occurs when **legitimate users cannot access systems or data**

Impact:

- Service outages
- Business disruption and financial loss
- Loss of customer trust
- Reputation damage
- Productivity loss

Main Causes:

- Denial of Service (DoS) attacks
- Hardware or software failure
- Network overload or bandwidth exhaustion
- Redundant arrangement failures
- Power outages

Example: DDoS Attack

Attackers flood servers with traffic, making legitimate users unable to access services

Summary: Key Takeaways

1. **Cybersecurity** protects software, hardware, and data resources online
2. The **CIA Triad** (Confidentiality, Integrity, Availability) is fundamental to information security
3. Cybersecurity involves **people, processes, and technology**
4. The market is growing rapidly ($213B in 2025)
5. **Cyberattacks** can breach confidentiality, availability, or integrity
6. Common threats include malware, phishing, DoS, ransomware, and more
7. **Education and awareness** are critical—humans are often the weakest link

Remember

Cybersecurity is a **continuous process**, not a one-time solution!

Thank You!

Questions?

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